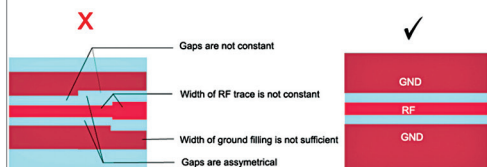
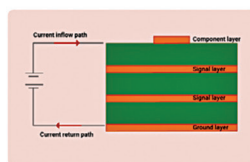
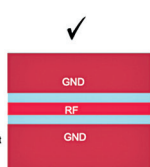


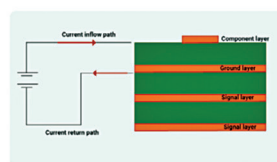
1. Choose appropriate transmission line – **Microstrip or Co-Planar Wave guide**
2. Trace should maintain **constant characteristic impedance** – Constant trace width
3. **Avoid sharp turns** and right-hand bends. Bend radius ≥ 3 times of trace width, Mitering
4. Dedicated **uninterrupted ground plane** for every layer with RF trace.
5. It is **not** encouraged to **route rf signal through vias** between different layers.
6. Choice of dielectric - **Low dielectric constant and loss tangent** – FR4 Derivatives (FR408HR)
7. Don't place any other traces close/parallel to RF trace – Mutual coupling.



Ref: Infineon RF Layout Guidelines - AN91445

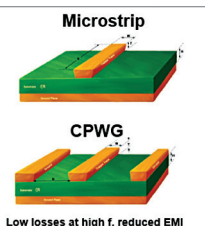


Incorrect Ground Plane Placement



Desired Ground Plane Placement

Ref: Sierra Circuits RF & Microwave Design Guide



Ref: Sierra Circuits RF & Microwave Design Guide

Another key parameter is the Q factor, or quality factor, which defines a component's efficiency in reactive circuits. For RF applications, Q values in the range of 3000 to 4000 are typical, making low-ESR components especially desirable.

In short, smart RF design is not just about picking

transmission line, or even an unintended antenna, in RF systems. This transition becomes critical when the wire length approaches a quarter of the signal's wavelength, triggering the need for distributed rather than lumped analysis.

In parallel, power supply noise emerges as a major challenge in high-frequency wireless design. Digital power rails are inherently noisy, and when these are connected to RF ICs, the resulting transients can disrupt frequency synthesis, especially at gigahertz-scale operations where even micro-second deviations can corrupt data transmission.

To mitigate this, power supply decoupling is essential. This typically involves placing a combination of capacitors in parallel close to the IC. High-frequency noise is handled using small ceramic capacitors (0.01-0.1 μ F), while larger electrolytic or tantalum capacitors (1-10 μ F) manage low-frequency noise and power surges. These capacitors also act as local charge reservoirs, stabilising the voltage during sudden current demands.

Component placement is just as critical as selection. Decoupling capacitors should be positioned as close as possible to the power pins. Following these practices is foundational to achieving signal integrity

and RF stability in compact, high-speed wireless designs.

Stage 3: Component selection: Inductors, capacitors

While much of the focus in wireless product development falls on MCUs, antennas, and wireless chipsets, passive components, such as capacitors, inductors, filters, baluns, diplexers, and duplexers, play an equally critical role in maintaining signal integrity and system stability at high frequencies.

In RF design, assuming ideal component behaviour is a costly mistake. Unlike their textbook equivalents, real-world components come with parasitic elements. Take capacitors, for example: beyond basic capacitance, they exhibit equivalent series resistance (ESR) and equivalent series inductance, stemming from their internal structure. These non-ideal characteristics distort expected performance, particularly as frequency increases.

A capacitor, ideally, should show decreasing impedance with rising frequency. But beyond a certain point, known as the self-resonant frequency (SRF), its behaviour reverses, and it begins acting more like an inductor. To avoid this, components must be operated well below their SRF to ensure predictable behaviour.

the right active components; it is about understanding the real-world limits of passive ones and designing around them with precision.

Stage 4: RF layout and grounding

Effective RF layout and grounding are fundamental to reliable high-frequency circuit performance. While standard PCB practices may suffice at lower frequencies, RF systems demand far tighter tolerances and greater attention to electromagnetic behaviour.

Microstrip lines, common in DC and low-frequency designs, begin to exhibit unacceptable energy losses at higher frequencies. To mitigate this, coplanar waveguides with ground structures are widely used. These feature a central signal trace flanked by grounded traces and isolated from the top layer by a dielectric substrate. For RF applications, traditional FR4 materials are often replaced with low-loss variants that offer reduced dielectric constant and minimal signal attenuation.

Maintaining consistent trace width and spacing is critical to preserving the characteristic impedance across the signal path. Variations can lead to impedance mismatches, introducing reflections and power loss. Sharp 90° bends should be avoided in favour of gradual curves